

Remote Assessment of Heart Rate by Skin Color Processing

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Abstract—Oxy-hemoglobin and deoxy-hemoglobin, main blood chromatophores flowing through the arteries and veins, respectively, reflect sunlight differently. The biggest difference is at a specific wavelength in the visible spectrum. The amount of blood in the arteries varies during cardiac activity. As a result, the skin color changes, but its limits cannot be noticed in heart cycle. The authors present an algorithm, for video signal processing, to achieve remote heartbeat having the following steps: selection of the region of interest, filtering using principal component analysis and empirical mode decomposition. The proposed method uses fewer pixels than those used by other methods presented in the literature and was confirmed by testing on subjects in the room light conditions by using a pulse oximeter.

Keywords—heart rate, facial expression, pattern recognition, computer vision, biomedical signal processing

I. INTRODUCTION

Medical analyses from the last two decades have shown a significant relationship between the autonomic nervous system and cardiovascular mortality. Heart Rate (HR) is an important parameter in many medical investigations [1] and it is used to measure irregular pulse rate. The post-processing of HR irregularities provides precious elements to the patient diagnosis. The evaluation process is more difficult for certain populations like premature neonates, infants at risk of sudden infant death syndrome, burn victims, and elderly, where the skin is fragile and damageable. In the same time, there is a growing interest in remote monitoring (i.e. outside of clinical environment) that can provide data, to caregivers (doctors, nurses, informal caregivers) about long-term evolution and statistical analyses [2]. Public compliance depends on the monitoring being contactless, non-obtrusive, and non-invasive.

Modern technologies are based on creating a ubiquitous info-communication environment. Non-contact, non-restrictive, and non-invasive sensing techniques have been researched and developed. They can use the cyclic movements of certain parts of body, color or temperature variation of the skin for monitoring heart beats.

In the literature the aspect of non-contact/non-invasive sensing has been addressed by a significant number of papers. Suzuki et al. [1] developed a system to monitor cardiac activity

using microwave Doppler radar. Da He [2] has described a continuous, wearable, and wireless heart monitor that is worn at the ear. It measures the ballistocardiographic motion of the head using a MEMS tri-axial accelerometer. Huang has investigated a noncontact HR monitoring using capacitive-coupled device integrated into a home sofa [3]. Electrodes were placed on the backrest of a sofa with only the chair covering by the user's clothing. The system is able to measure contactless HR through thin cotton clothing. Parameters were saved on a cloud database, providing users with a long-term monitoring for physiological information. Some research evaluates heart rate through thermal image processing of human face [4]. The heart rate is represented as a function of skin small temperature variation during pulsation. Pavlidis et al. [5, 6] took the thermal data from a cooled infrared imaging system and used the Fast Fourier transform to filter data. Temperature information cannot be extracted from images of the uncooled infrared cameras because of their lower temperature sensitivity. Jing and Haiyun proposed [7] a head movement detection algorithm based on the centroid coordinate change of IR sequences.

Video based measurements of subtle head oscillations accompanying the cardiac cycle have been studied by Balakrishnan [8]. He performed the principal component analysis to decompose the trajectories into a set of mono-dimensional component motions, and identified the picks of the most periodic component for HR.

Takano and Ohta reported the results of using the brightness data from a time-lapse image of a CCD camera [9]. The data were measured for 30 s. Chen et al. [10] investigating the Hilbert-Huang Transform of the face reflectance for the HR measurement. The authors decomposed face reflectance from RGB, selected green channel and applied empirical mode decomposition to separate HR from the rest of components including noise.

In our proposed paper, we have selected a signal corresponding to a certain wavelength where there is the highest difference of skin color during a cardiac cycle. The nonlinear phenomena associated with HR are determined by complex interactions of electrophysiological, hemodynamic, humoral variable, central and autonomic nervous regulations. Therefore, nonlinear methods have to be used for investigation these biomedical signals.

II. MATERIALS AND METHODS

Skin light scattering and absorption properties are the result of the various skin chromatophores and the reflective index fluctuation on a microscopic level. Hemoglobin, the most important chromophore in the blood, carries oxygen to all human cells. There are two components forms: oxy-hemoglobin, having a higher affinity for oxygen, and deoxy-hemoglobin that has a higher affinity for carbon dioxide. The pressure and the oxygen concentration of the blood are variable during heart cycle. A short hemodynamic investigation is sufficient to perform the heart rate measurement Heart Rate monitoring has evolved rapidly during last three decades [10].

The pixels values from the original image of face are directly proportional to the reflectance. Much research has been done to measure the diffuse reflectance spectra from the original image [11, 12]. Indirect information about the reflectance can be obtained if absorption spectrum is known [13,14]. Absorption spectra of melanin and hemoglobin are shown in Fig. 1, from [12, 15]. The absorption spectrum of hemoglobin has two local maxima. One of them is in short wavelength range, both oxy-hemoglobin and deoxy-hemoglobin and the other one is around the 550 nm range with two or one maxima, respectively. The biggest difference between the spectra of the two components of hemoglobin, oxygenated and deoxygenated, occurs at about 560 nm, Fig. 1. A similar difference may be found also in reflectance spectra. The amount of blood that highlights this difference varies over time due to the pressure generated by the heartbeat. Video surveillance can be used to obtain information on heart rate and its variability.

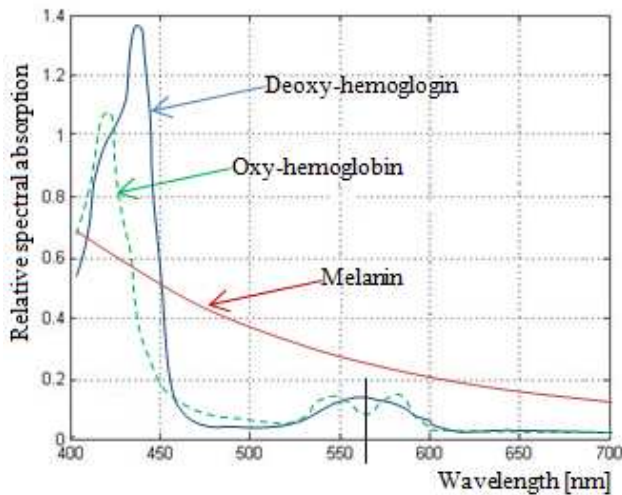


Fig. 1. Relative spectral absorption of melanin, oxy-hemoglobin, and deoxy-hemoglobin in the visible spectrum (modified reproduction from [12])

The proposed method tracks the features of the head image, and signals were extracted from the subject with a frontal view of the face. Signal processing should be used because the skin color has a small variation that is very hard to be noticed, Fig. 2. Principal component analysis (PCA) was designed performed to decompose the complex signal into a set of mono-dimensional components. PCA transforms a number of

possibly correlated variables into a smaller number of variables called principal components [16, 17].

The first step of PCA is to identify a new set of orthogonal coordinate vectors through the data in which to re-express them. This is achieved by finding the directions of a certain variance through the coordinates in the n -dimensional space on which to project the position time-series to obtain the signal.

The n values measured, during each frame, generate a matrix X with $n \times m$ dimensions. Each variable has a row vector; each of them contains all samples obtained during temporal variation. This matrix is linearly transformed into another matrix, Y , having the same dimension as the initial one by using a new $n \times n$ matrix, P :

$$Y = PX \quad (1)$$

Equation (1) represents a general form of basis change for the initial data from X matrix. The original data, X is being projected on to the column of P . The rows of P are a new basis for representing the columns of X , and could become principal component direction if they are independent. PCA seeks to de-correlate the original data by finding the direction in which variance is maximized.

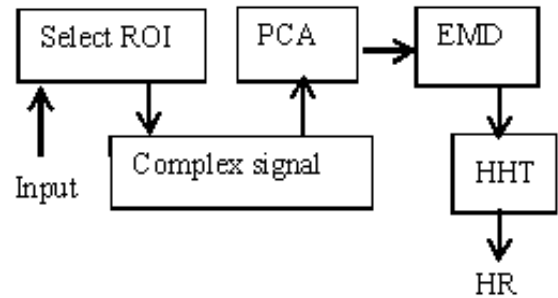


Fig. 2. Signal processing for HR evaluation

In the second step, the temporal signals are processed one by one. The investigation methods are different when the signals are stationary or non-stationary. Traditional Fourier analysis performs well when the system is linear and data are periodic or stationary. Nevertheless, local non-linearity and different frequencies appear and cannot be analyzed by means of the basic functions of Fourier Transform, sin/cos, because they are global.

Like wavelet or Fourier transforms, recently, Huang proposed the Empirical Mode Decomposition method, EMD based into a set of basic functions [18, 19]. These functions are derived from the signal itself and capture the repeating behavior at the same particular time scale. They are based on instantaneous frequencies by identifying the intrinsic mode functions, IMF by their characteristic time scale. Nearly any signal can be decomposed into a finite set of functions, whose Hilbert transforms give physical instantaneous values by their characteristic time scales from the data empirically [20].

The EMD calculation is a cascade of sifting operations. First sifting computes a bloc of the first IMF and a block of corresponding residue. This residue is then used as the input to the second sifting and so on [15].

A sum of two harmonic signals is shown on Fig. 3a. The local minima and maxima represent upper and lower envelope of the signals. The upper envelope, red color in Fig. 3a, is represented by the cubic spline line that connects all the local maxima. Similarly, the lower envelope, green color in Fig. 3a, is represented by the cubic spline line that connects all the local minima. Then, the mean value of the two harmonic signal sum, $m_1(t)$ is evaluated. The first component of the first sift, $h_1(t)$ is obtained as follow:

$$h_1(t) = X(t) - m_1(t) \quad (2)$$

This component could be an IMF if it satisfies two conditions [15]: in the whole data set the number of extremes and the number of zero crossing must either be equal or differ at most by one; the mean value of the envelope defined by the local maxima and the envelope defined by local minima is zero at any point. Otherwise, $h_1(t)$ is treated as data, then:

$$h_{11}(t) = h_1(t) - m_{11}(t) \quad (3)$$

This is a sifting process, and is repeated k times until $h_{1k}(t)$ meets two conditions prior. The sifting process has to eliminate riding waves and to make the wave profiles more symmetric.

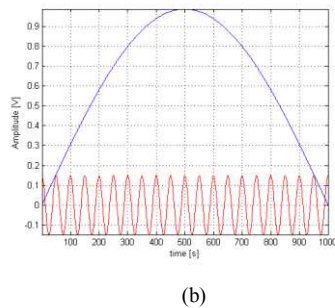
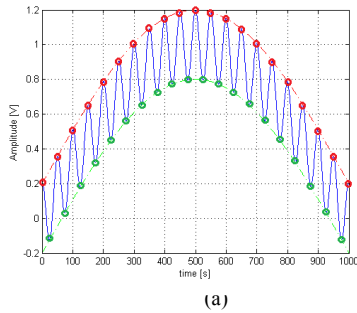


Fig. 3. Illustration of EMD decomposition; (a) the complex signal in thin solid line, with the upper and lower envelopes in dot-dashed lines; (b) the decomposed signal in IMF - red color, and residue in blue one.

The IMF h_{1k} contains the finest scale or the shortest period component of the signal. The residue will be obtained after it is separated from the signal:

$$r_1(t) = X(t) - h_{1k}(t) \quad (4)$$

A stop criterion for sifting process is applied in order to guaranty that the IMF components retain enough physical sense of both amplitude and frequency modulation. The threshold for the standard deviation between the present and previous iteration [15]:

$$SD = \sum_{t=0}^T \left[\frac{h_{1(k-1)}(t) - h_{1k}(t)}{h_{1(k-1)}^2(t)} \right]^2 \quad (5)$$

Optimal value should be between 0.2 and 0.3, most of the time [3]. After the all IMFs are subtracted, the initial signal will be obtained by summing up the all IMFs with the final residue:

$$X(t) = \sum_{i=1}^n h_i + r_n \quad (6)$$

The Hilbert-Huang transform (HHT) has been used to analyses the input signal by decomposition it into a number of cyclic components using EMD. Given an arbitrary series $X(t)$, the analytical signal $Z(t)$ is:

$$Z(t) = X(t) + jY(t) = a(t) \exp(j\theta(t)) \quad (7)$$

where $Y(t)$ is Hilbert Transform of $X(t)$ [3]:

$$Y(t) = \frac{1}{\pi} PV \int_{-\infty}^{\infty} \frac{X(\tau)}{t - \tau} d\tau \quad (8)$$

and PV indicates the Cauchy principal value. The instantaneous frequency of each IMF is defined as [12, 13]:

$$\omega(t) = d\theta(t)/dt \quad (9)$$

The experimental setup consisted of a video camera and a finger pulse oximeter. The streaming video frame rate was 30 fps. The images, with the video resolution of 576x720 pixels, were saved in AVI video format without compression for 20 s. The analysis was performed for a 40x120 pixels rectangular region of interest (ROI), on the forehead area, by using the Viola Jones face detector from OpenCV 2.4.10 library in MATLAB [16]. For the three channels R, G, and B, a single channel, $I(x,y)$, is formed by using the following relation:

$$I(x,y) = 0.254 * R(x,y) + 0.659 * G(x,y) + 0.087 * B(x,y) \quad (10)$$

This bi-dimensional signal from each video signal frame is a combination of the three color channels. It corresponds to the wavelength of which the two hemoglobin components can be detected during systole and diastole [14].

The signal from the ROI of first frame is shown in Fig. 4. In total there are 600 frames. The signal appears to be random, without periodic components. The first significant results appear after the PCA signal processing phase. Information on heart rate is contained in the temporal sequence of 600 frames. Data for the PCA analysis were the 120 values of the pixels from ROI row recorded during 600 successive time moments. All 40 columns have been processed.

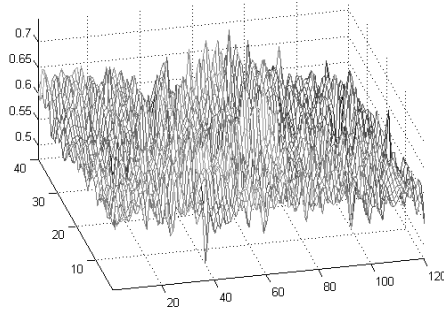


Fig. 4. 3D representation of ROI for the first frame

The influence of the order of the principal direction can be noticed on Fig. 5. A lot of noise is presented on the first principal direction, and it decreases as the order of the principal direction increases. At the same time, periodic component is something more readable.

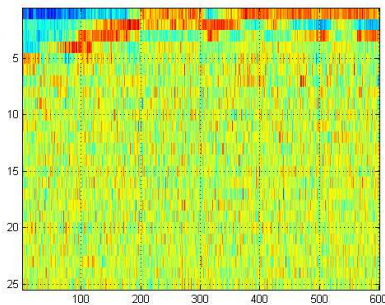


Fig. 5. Initial ROI row projected on 25 principal directions

A total of 25 principal directions were selected for each column of initial data. 25 out of 120 temporal evolutions have the largest variances and are shown in Fig.5. The initial image projected onto initial basis of the ROI pixel values, after the PCA phase, is presented on Fig. 6.

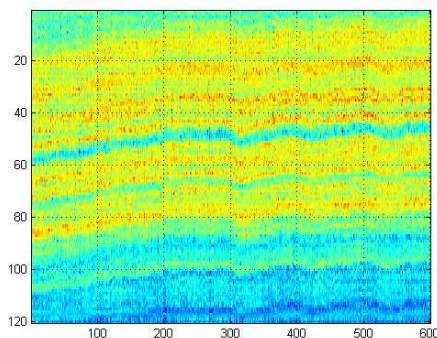


Fig. 6. Projected image onto initial basis after PCA

Eigenvalues were arranged in descending order. The best results were obtained by applying the proposed method using any eigenvalue between the third up to the ninth. Initial data were projected on the correspondent eigenvectors, and the second filtering method, EMD, was used. The first three IMFs are shown in Fig. 8. They were obtained by projecting data on the 4th principal direction assessed by the previous method.

III. DISCUSSION

Information about HR was taken in two ways: remote by using a camera, and local by a pulse oximeter placed on a finger. In the first case, a rectangular area on the forehead was selected as ROI and the video signal was processed in two steps: by PCA and EMD.

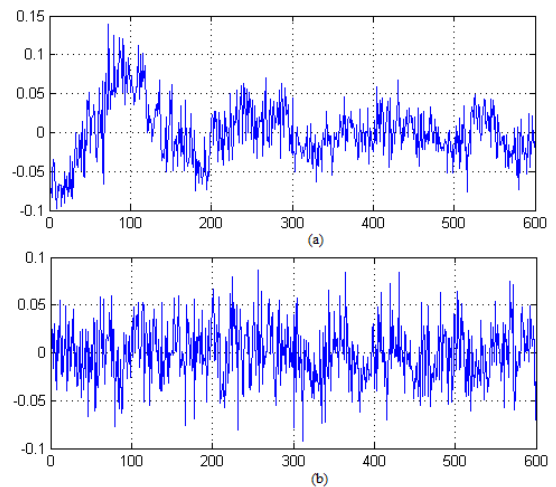


Fig. 7. Mono-dimensional data projected on the 1st (a) and 4th (b) principal direction

HR for the same investigation period of 20 seconds was taken from the video capture. Using PCA analysis showed that this method can compress data from 120 directions to 25 without losing significant information. At the same time, similar data were obtained from any of the 40 columns of three-dimensional matrix of the input video data. As a result, data projections on 25 principal directions were introduced successively in the next step of processing. Data projections on the first two principal directions have high value of noise component, and the projections corresponding to higher orders of eigenvalues than 9 contain low levels of signal. Projection on the fourth principal direction provides a better detection of the frequency bands associated with IMF2, Fig. 8.

The finger pulse oximeter signal was collected while the video was been recorded, Fig. 9a.

The final results of this investigation can be analyzed by using the data shown in Fig. 8 and Fig. 9. All 40 columns of ROI were processed by the same algorithm, and the results were similar.

Fourier transform of data from the finger pulse oximeter shows a value of 67.2 beats per minute for HR, a constant value over this investigation period of 20 seconds, Fig. 9b. This is not completely true because the pulse oximeter values change over time. The FFT curve is not narrow, which indicates the existence of spectral components with close values, but with smaller amplitude.

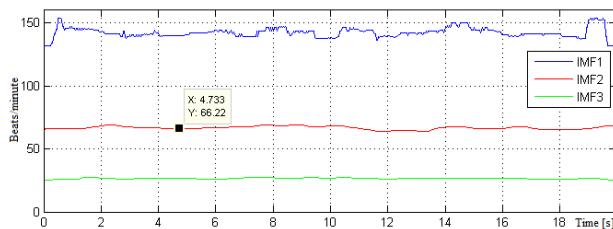


Fig. 8. The first three IMFs for data projected on the 4th principal direction

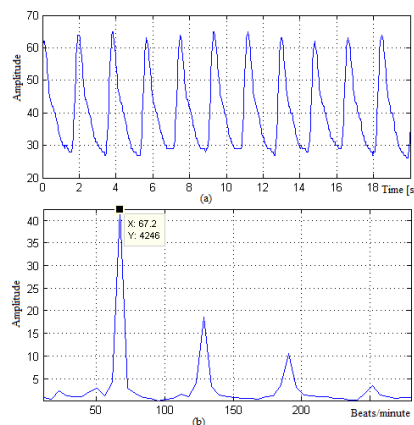


Fig. 9. Output signal from pulse oximeter (a) and its Fourier transform (b)

HR values that can be seen on IMF2 are close to those given by the pulse oximeter. A variation of HR between 66.1 and 67.8 beats per minute is noticed during the 20 seconds of capture.

IV. CONCLUSIONS

In this paper we present a new remote assessment for HR. Input is the pixel intensity from forehead area corresponding to a certain wavelength where an appreciable difference of oxy-hemoglobin and deoxy-hemoglobin contributions exists. Summing weighted of the pixel intensities of the R, G and B channels by choosing appropriate wavelengths of 560 nm, and processing in two steps are sufficient for remote evaluation of HR. Temporal values from 120 pixels during 600 successive time moments were sufficient in the first of the processing step.

After that, in the second stage they were reduced to the projection onto one main direction. We have applied PCA and EMD to the pixel sequence, and have obtained the same results as traditional devices (i.e. pulse oximeter). The proposed method consists in processing a smaller number of pixels in two stages, PCA and EMD. HR evaluations are presented in the literature from processing all the pixels in ROI in one stage, PCA or EMD. Results are similar.

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